Day – 5 Inter-VLAN Routing Concepts??? Theory.

Task – 1Create VLANs.

- Step 1: Setting up the Network Topology in Packet Tracer
- To configure VLANs on a single switch, build the following layout:
- Place one 2960 Switch (or similar managed switch model).
- Place three PCs on your workspace: PC 0 (Insta), PC 1 (Zomato), and PC 2 (Flipkart).
- Connect the PCs to the switch using Copper Straight-Through cables:
- PC 0 to switch port FastEthernet 0/1
- PC 1 to switch port FastEthernet 0/2
- PC 2 to switch port FastEthernet 0/3
- Step 2: Defining the VLANs and IP Addressing Scheme
- Each VLAN requires its own distinct broadcast domain (subnet) so devices can be logically segmented:
- VLAN 10 (Insta): Subnet 192.168.10.0/24
- PC 0 IP Address: 192.168.10.10 (Subnet Mask: 255.255.255.0)
- VLAN 20 (Zomato): Subnet 192.168.20.0/24
- PC 1 IP Address: 192.168.20.10 (Subnet Mask: 255.255.255.0)
- VLAN 30 (Flipkart): Subnet 192.168.30.0/24
- PC 2 IP Address: 192.168.30.10 (Subnet Mask: 255.255.255.0)
- Step 3: Configuring the PCs
- Configure the IP address, subnet mask, and leave the default gateway blank for now (since there is no router configured yet for inter-VLAN routing):
- Click PC 0 -> Desktop -> IP Configuration -> Set IP to 192.168.10.10 and Mask to 255.255.255.0.
- Click PC 1 -> Desktop -> IP Configuration -> Set IP to 192.168.20.10 and Mask to 255.255.255.0.
- Click PC 2 -> Desktop -> IP Configuration -> Set IP to 192.168.30.10 and Mask to 255.255.255.0.
- Step 4: Creating VLANs and Assigning Ports on the Switch
- Click on the switch, open the CLI tab, and enter the following commands to create the three VLANs and assign their respective ports:
- Switch> enable
- Switch# configure terminal
- ! Create VLANs and assign names
- Switch(config)# vlan 10
- Switch(config-vlan)# name Insta

- Switch(config-vlan)# exit
- Switch(config)# vlan 20
- Switch(config-vlan)# name Zomato
- Switch(config-vlan)# exit
- Switch(config)# vlan 30
- Switch(config-vlan)# name Flipkart
- Switch(config-vlan)# exit
- ! Assign FastEthernet 0/1 to VLAN 10 (Insta)
- Switch(config)# interface fastEthernet 0/1
- Switch(config-if)# switchport mode access
- Switch(config-if)# switchport access vlan 10
- Switch(config-if)# exit
- ! Assign FastEthernet 0/2 to VLAN 20 (Zomato)
- Switch(config)# interface fastEthernet 0/2
- Switch(config-if)# switchport mode access
- Switch(config-if)# switchport access vlan 20
- Switch(config-if)# exit
- ! Assign FastEthernet 0/3 to VLAN 30 (Flipkart)
- Switch(config)# interface fastEthernet 0/3
- Switch(config-if)# switchport mode access
- Switch(config-if)# switchport access vlan 30
- Switch(config-if)# exit
- Switch(config)# end
- Switch# write memory
- Step 5: Verifying the VLAN Configuration
- To confirm that the VLANs have been successfully created and the ports assigned correctly, run the verification command on the switch CLI:
- Switch# show vlan brief
- Expected Output:
- You should see a summary table listing your custom VLAN names and their assigned ports:
- VLAN Name Status Ports
- ---- ---------- --------- ------------------------------
- 1 default active Fa0/4, Fa0/5, Fa0/6, ...
- 10 Insta active Fa0/1
- 20 Zomato active Fa0/2
- 30 Flipkart active Fa0/3

Task – 2 Config router on the stick.

- To enable communication between the isolated VLANs (Insta, Zomato, and Flipkart) created earlier, you need to implement Router-on-a-Stick inter-VLAN routing. This involves connecting a router to the switch via a single physical trunk link and configuring sub-interfaces for each VLAN.
- Step 1: Connecting the Router to the Switch
- Place a router (e.g., Cisco 1941 or 2911) in your Packet Tracer workspace.
- Connect GigabitEthernet 0/0 on the router to GigabitEthernet 0/1 (or any available gigabit uplink port) on your switch using a Copper Straight-Through or Cross-Over cable.
- Step 2: Configuring the Switch Trunk Port
- Before configuring the router sub-interfaces, you must configure the switch port connected to the router as an 802.1Q trunk port so it can carry traffic for all VLANs.
- Open the switch CLI and enter the following commands:
- Switch> enable
- Switch# configure terminal
- Switch(config)# interface gigabitEthernet 0/1
- Switch(config-if)# switchport mode trunk
- Switch(config-if)# no shutdown
- Switch(config-if)# exit
- Switch# write memory
- Step 3: Configuring Router Sub-Interfaces
- Open the router CLI, enable the physical interface, and create a sub-interface for each VLAN. You will encapsulate the traffic using IEEE 802.1Q (encapsulation dot1Q <VLAN_ID>) and assign the default gateway IP address for each subnet:
- Router> enable
- Router# configure terminal
- ! Enable the main physical interface without assigning an IP address
- Router(config)# interface gigabitEthernet 0/0
- Router(config-if)# no shutdown
- Router(config-if)# exit
- ! Configure Sub-interface for VLAN 10 (Insta)
- Router(config)# interface gigabitEthernet 0/0.10
- Router(config-subif)# encapsulation dot1Q 10
- Router(config-subif)# ip address 192.168.10.1 255.255.255.0
- Router(config-subif)# exit
- ! Configure Sub-interface for VLAN 20 (Zomato)
- Router(config)# interface gigabitEthernet 0/0.20
- Router(config-subif)# encapsulation dot1Q 20
- Router(config-subif)# ip address 192.168.20.1 255.255.255.0
- Router(config-subif)# exit

- ! Configure Sub-interface for VLAN 30 (Flipkart)
- Router(config)# interface gigabitEthernet 0/0.30
- Router(config-subif)# encapsulation dot1Q 30
- Router(config-subif)# ip address 192.168.30.1 255.255.255.0
- Router(config-subif)# exit
- Router(config)# end
- Router# write memory
- Step 4: Updating PC Default Gateways
- Since the router is now routing traffic between the subnets, update the IP configuration on your end devices to include their respective default gateways:
- PC 0 (Instent / VLAN 10):
- IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1
- PC 1 (Zomato / VLAN 20):
- IP Address: 192.168.20.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.20.1
- PC 2 (Flipkart / VLAN 30):
- IP Address: 192.168.30.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.30.1

Task – 3 Assign the switch port to connected to router.

- To ensure that traffic from all three VLANs (Insta, Zomato, and Flipkart) can flow between the switch and the router via the single physical link, configure the switch port as a trunk port and verify its operational status.
- Step 1: Configuring the Trunk Port on the Switch
- Assuming the switch's GigabitEthernet 0/1 port is connected to the router's physical interface, enter the following commands in the switch CLI:
- Switch> enable
- Switch# configure terminal
- Switch(config)# interface gigabitEthernet 0/1
- Switch(config-if)# switchport mode trunk
- Switch(config-if)# no shutdown
- Switch(config-if)# exit
- Switch(config)# end
- Switch# write memory
- Step 2: Verifying Trunking Status
- To confirm that the trunk link has formed successfully and is actively carrying traffic across your VLANs, run the verification command on the switch:

- Switch# show interfaces trunk
- What to Look For in the Output:
- You should see a status table indicating that the port is operating as a trunk and listing the permitted/active VLANs:
- Port Mode Encapsulation Status Native vlan
- Gi0/1 on 802.1q trunking 1
-
- Port Vlans allowed on trunk
- Gi0/1 1-1005
-
- Port Vlans allowed and active in management domain
- Gi0/1 1,10,20,30

Task – 4 Test inter – VLAN connectivity.

- Step 1: Executing the Ping Test
- Open PC 0 (in the Insta VLAN / subnet 192.168.10.10).
- Go to the Desktop > Command Prompt tab.
- Ping PC 1 (in the Zomato VLAN / subnet 192.168.20.10):
- C:\> ping 192.168.20.10
- You will receive successful echo replies (Reply from 192.168.20.10: bytes=32 time<1ms TTL=127).
- Step 2: Why the Ping Succeeds
- The ping succeeds because you have successfully established Router-on-a-Stick inter-VLAN routing using 802.1Q trunking. Here is the step-by-step path the packet takes:
- Local Gateway Lookup: PC 0 wants to reach an IP address (192.168.20.10) that is outside its own subnet (192.168.10.0/24). Because of this, PC 0 wraps the packet in an Ethernet frame addressed to its configured default gateway (192.168.10.1, which is the router's sub-interface).
- Trunk Tagging: The switch receives the frame on access port Fa0/1 (assigned to VLAN 10), realizes it needs to go to the router via the trunk port (Gi0/1), and tags the frame with an 802.1Q header for VLAN 10.
- Router Sub-Interface Decapsulation: The router receives the tagged frame on its physical interface (Gi0/0), passes it to sub-interface GigabitEthernet 0/0.10, strips the tag, and processes the packet in Layer 3.
- Routing Decision: The router checks its routing table and sees that the destination subnet (192.168.20.0/24) is directly connected via sub-interface GigabitEthernet 0/0.20.

- Encapsulation & Delivery: The router encapsulates the packet for VLAN 20, tags it with 802.1Q ID 20, and sends it back down the trunk link to the switch. The switch strips the tag and forwards the frame out access port Fa0/2 to PC 1.
- Because the switch port connected to the router is configured as a trunk (switchport mode trunk), the router has active 802.1Q sub-interfaces (.10, .20), and all devices have correct default gateways pointing to those sub-interfaces, communication between isolated VLANs flows seamlessly.

Task – 5 Refactor your router on the stick.

- 1. On the Switch: Create the VLAN and assign a port
- Tell the switch about the new VLAN and plug your Spotify PC into a port assigned to it:
- Switch(config)# vlan 40
- Switch(config-vlan)# name Spotify
- Switch(config-vlan)# exit
- Switch(config)# interface fastEthernet 0/4
- Switch(config-if)# switchport mode access
- Switch(config-if)# switchport access vlan 40
- 2. On the Router: Create a sub-interface
- Give the router a gateway IP for this new VLAN so it can talk to Insta, Zomato, and Flipkart:
- Router(config)# interface gigabitEthernet 0/0.40
- Router(config-subif)# encapsulation dot1Q 40
- Router(config-subif)# ip address 192.168.40.1 255.255.255.0